

Energy Production Prediction Project Presentation

**Presented By :
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Overview

- Introduction
- Abstract
- Methodology
- Steps Performed
- Data Extraction
- Data Visualization
- Decomposition of Data
- Models:- MAE ,MSE ,RMSE , Decision Tree ,Random Forest ,Linear Regression,
- Conclusion

Introduction

- Combined-cycle power plants represent an efficient approach to energy generation by combining gas turbines, steam turbines, and heat recovery steam generators within a single cycle.
- This system capitalizes on the high-efficiency potential of using exhaust gases from gas turbines to power steam turbines, thereby increasing overall energy production.
- The primary objective of this project is to develop a predictive model for energy output as a function of specific ambient conditions and exhaust vacuum levels.
- By analyzing these factors, we aim to provide actionable insights for optimizing plant operations and maximizing energy efficiency.

Abstract

This project develops a predictive model for energy output in a combined-cycle power plant, utilizing ambient and operational data to optimize performance. Combined-cycle power plants generate electricity efficiently by harnessing both gas and steam turbines, where exhaust heat from gas turbines powers steam turbines in a continuous cycle.

The model leverages 9,568 observations over six years, focusing on key variables: temperature, exhaust vacuum, ambient pressure, and relative humidity. By quantifying how these factors influence energy production, this model provides insights that allow operators to make real-time adjustments for improved efficiency.

- Goal: Enable data-driven decisions that maximize energy output and optimize resource management, enhancing overall plant productivity.

Methodology

Training and Testing were both done in Jupyter Notebook, Google Collab which provides a Jupyter notebook environment and free RAM and GPU to train and test the models.

Programming Language:- Python

Libraries:- Pandas, NumPy, Streamlit, Joblib, Plotly (and by extension, scikit-learn)(Optional: Seaborn might be used or considered for additional visualizations.)

Algorithm:- MAE, MSE, RMSE, Decision Tree, Random Forest, Linear Regression



Steps performed

- 1) Importing Data
- 2) Split the Data into Training & Test sets
- 3) Creating & Training the Model
- 4) Making Predictions
- 5) Evaluating & Improving Predictions
- 6) Streamlit App Development

Dataset extraction

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

#reading the file

```
file_path = '/Altamash/Excelr code/PROJECTS/PROJECT_ ENERGY_PRODUCTION/Copy of Regrerssion_energy_produ
df = pd.read_csv(file_path, delimiter=';')
```

#In Exploratory Data Analysis (EDA), the head() function is used to display the first few rows of a data
`df.head()`

	temperature	exhaust_vacuum	amb_pressure	r_humidity	energy_production
0	9.59	38.56	1017.01	60.10	481.30
1	12.04	42.34	1019.72	94.67	465.36
2	13.87	45.08	1024.42	81.69	465.48
3	13.72	54.30	1017.89	79.08	467.05
4	15.14	49.64	1023.78	75.00	463.58

#In Exploratory Data Analysis (EDA), the tail() function is used to display the last few rows of a data
`df.tail()`

Continue...

	temperature	exhaust_vacuum	amb_pressure	r_humidity	energy_production
0	9.59	38.56	1017.01	60.10	481.30
1	12.04	42.34	1019.72	94.67	465.36
2	13.87	45.08	1024.42	81.69	465.48
3	13.72	54.30	1017.89	79.08	467.05
4	15.14	49.64	1023.78	75.00	463.58

```
#to get no.of rows and columns  
df.shape
```

```
(9568, 5)
```

```
#to get basic info  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 9568 entries, 0 to 9567  
Data columns (total 5 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   temperature           9568 non-null   float64  
1   exhaust_vacuum         9568 non-null   float64  
2   amb_pressure           9568 non-null   float64  
3   r_humidity             9568 non-null   float64  
4   energy_production      9568 non-null   float64  
dtypes: float64(5)  
memory usage: 373.9 KB
```


Removing duplicate values

```
#checking for missing values  
df.isnull().sum()
```

```
temperature      0  
exhaust_vacuum    0  
amb_pressure      0  
r_humidity        0  
energy_production 0  
dtype: int64
```

No missing values in the dataset

```
#checking for duplicates values  
df.duplicated().sum()
```

```
41
```

41 duplicates in the dataset

```
#remove duplicated from the dataset  
df.drop_duplicates(inplace=True)
```

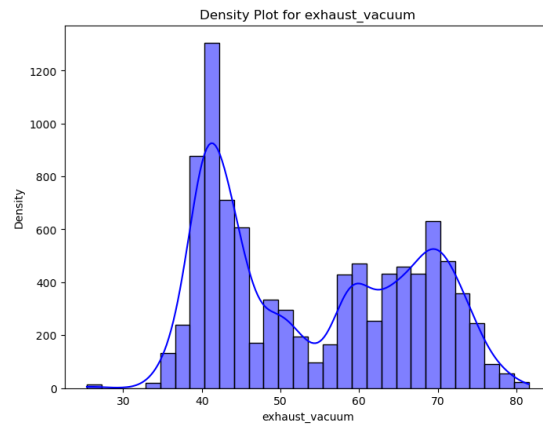
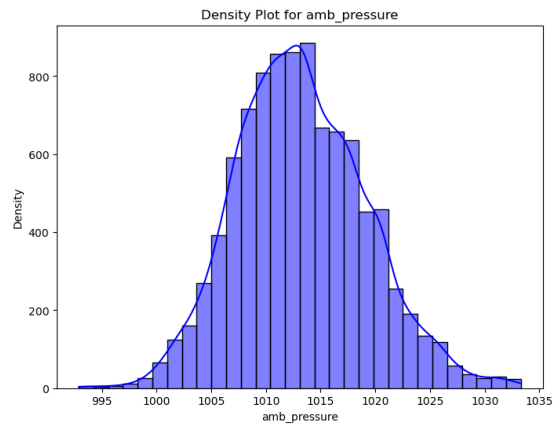
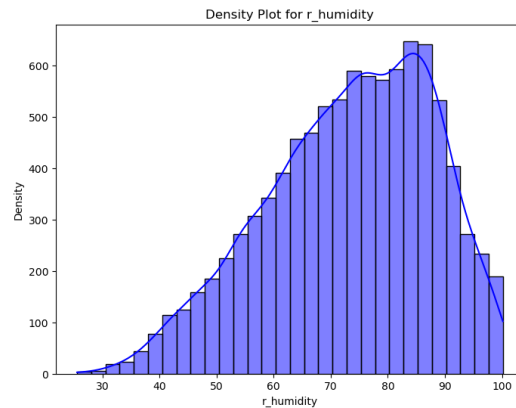
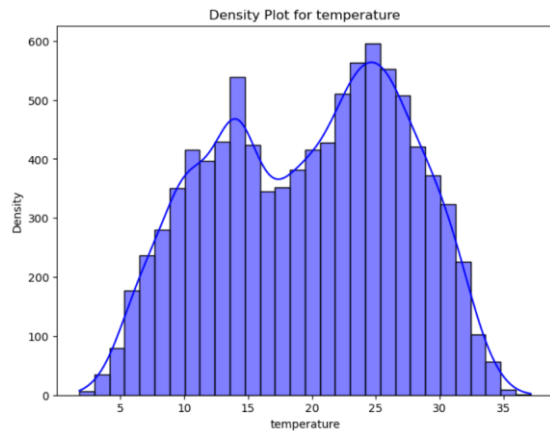
```
#recheck for duplicates in the dataset  
df.duplicated().sum()
```

```
0
```

Data visualization

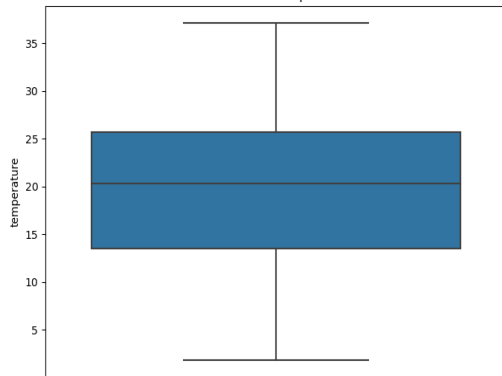
- Histogram: Shows data distribution by frequency across intervals.
- Box Plot: Displays data spread, center, and outliers.
- Pairplot: Visualizes relationships between pairs of variables.
- Correlation Matrix: Shows correlation values between variables.
- Scatter Plot: Plots relationships between two continuous variables.

Histogram

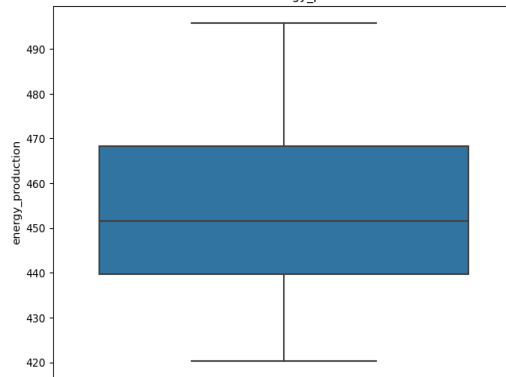


Box plot

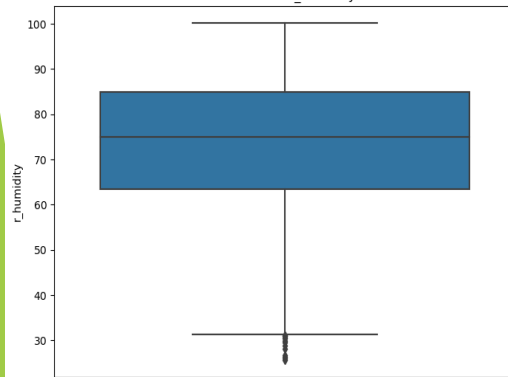
Box Plot for temperature



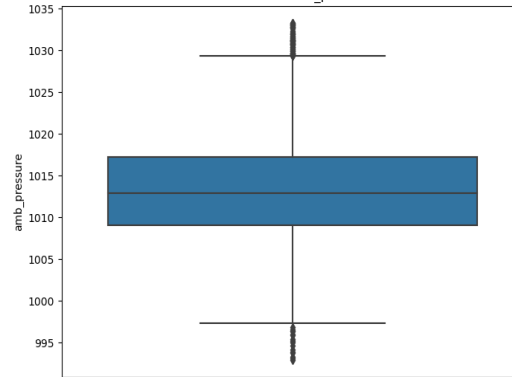
Box Plot for energy_production



Box Plot for r_humidity

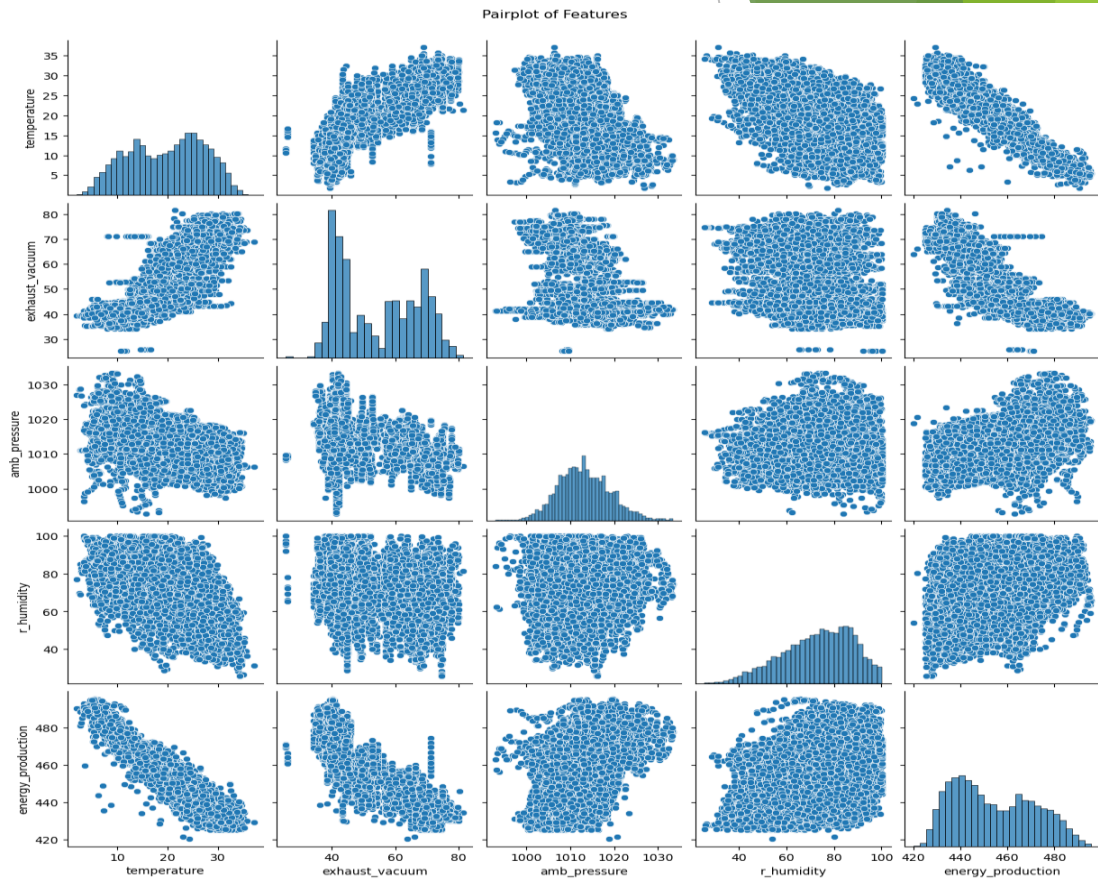


Box Plot for amb_pressure



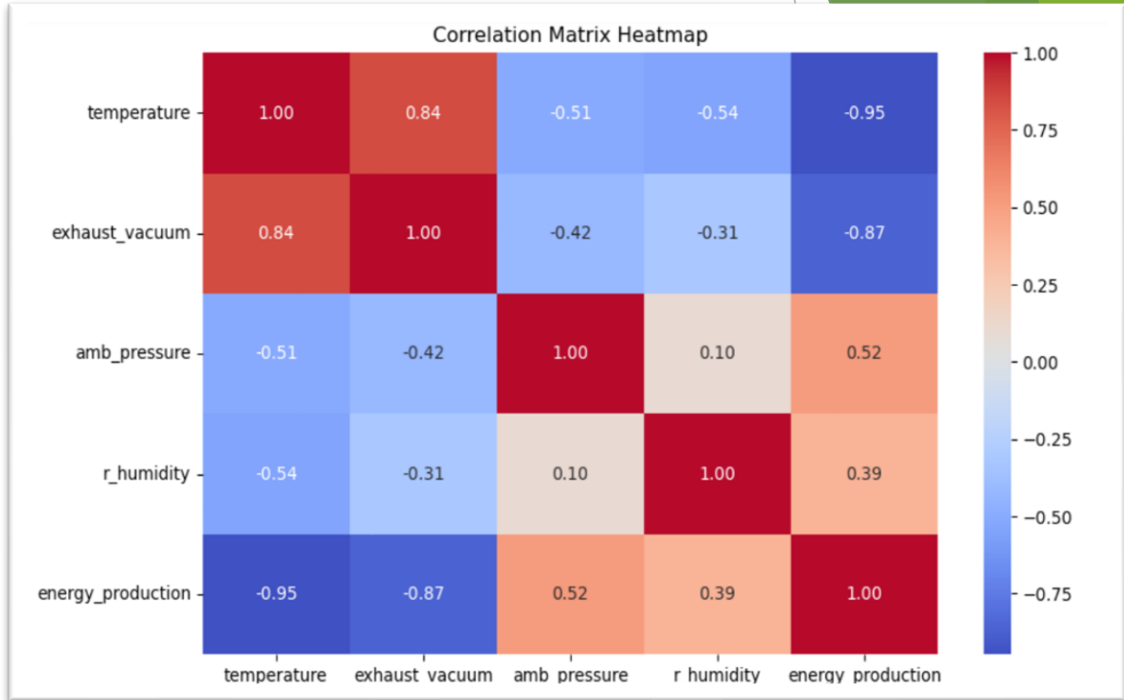
Pairplots

- This pairplot shows the relationships between several variables in a dataset.
- Diagonal plots show the distribution of each variable, while off-diagonal scatter plots reveal interactions between pairs of variables.
- Notably, temperature and energy production have a negative correlation, and there's a pattern between exhaust vacuum and energy production, suggesting these may be important relationships for further analysis.
- Temperature and energy production go in opposite directions: when one goes up, the other goes down.
- Exhaust vacuum and energy production seem linked in a clear pattern.



Correlation Matrix

- Temperature and energy production:
Strong negative (-0.95), meaning higher temperatures reduce energy production.
- Exhaust vacuum and energy production:
Strong negative (-0.87), higher vacuum reduces energy production.
- Ambient pressure and energy production:
Moderate positive (0.52), higher pressure increases energy production.
- Temperature and exhaust vacuum: Strong positive (0.84), both increase together.
- Relative humidity and energy production:
Weak positive (0.39), slight increase in energy with higher humidity.



Train and Evaluate Different Regression Models

The models listed are:

Regression Models

Linear Regression

Ridge Regression

Lasso Regression

Ensemble Methods

Random Forest

XGBoost

Random Forest

Decision Tree

Bagging Regressor

XGBoost

AdaBoost

Gradient Boosting

K-Nearest Neighbors

Continue....

Elastic Net

Model performance for Training set

- r2_score: 0.9279
- mean_squared_error: 4.5722
- mean_absolute_error: 3.6418

Model performance for Test set

- r2_score: 0.9278
- mean_squared_error: 4.5850
- mean_absolute_error: 3.6635

Linear Regression

Model performance for Training set

- r2_score: 0.9284
- mean_squared_error: 4.5561
- mean_absolute_error: 3.6213

Model performance for Test set

- r2_score: 0.9283
- mean_squared_error: 4.5693
- mean_absolute_error: 3.6441

Random Forest

Model performance for Training set

- r2_score: 0.9945
- mean_squared_error: 1.2590
- mean_absolute_error: 0.8852

Model performance for Test set

- r2_score: 0.9617
- mean_squared_error: 3.3396
- mean_absolute_error: 2.3406

Decision Tree

Model performance for Training set

- r2_score: 1.0000
- mean_squared_error: 0.0000
- mean_absolute_error: 0.0000

Model performance for Test set

- r2_score: 0.9327
- mean_squared_error: 4.4272
- mean_absolute_error: 2.9695

Bagging Regressor

Model performance for Training set

- r2_score: 0.9923
- mean_squared_error: 1.4974
- mean_absolute_error: 0.9987

Model performance for Test set

- r2_score: 0.9588
- mean_squared_error: 3.4648
- mean_absolute_error: 2.4380

XGBoost

Model performance for Training set

- r2_score: 0.9878
- mean_squared_error: 1.8781
- mean_absolute_error: 1.3643

Model performance for Test set

- r2_score: 0.9647
- mean_squared_error: 3.2042
- mean_absolute_error: 2.2316

Continue....

Ridge Regression

=====

Model performance for Training set

- r2_score: 0.9284
 - mean_squared_error: 4.5561
 - mean_absolute_error: 3.6213
-

Model performance for Test set

- r2_score: 0.9283
 - mean_squared_error: 4.5693
 - mean_absolute_error: 3.6441
- =====

K-Nearest Neighbors

=====

Model performance for Training set

- r2_score: 0.9647
 - mean_squared_error: 3.2003
 - mean_absolute_error: 2.3378
-

Model performance for Test set

- r2_score: 0.9468
 - mean_squared_error: 3.9368
 - mean_absolute_error: 2.8719
- =====

Gradient Boosting

=====

Model performance for Training set

- r2_score: 0.9539
 - mean_squared_error: 3.6556
 - mean_absolute_error: 2.8113
-

Model performance for Test set

- r2_score: 0.9461
 - mean_squared_error: 3.9614
 - mean_absolute_error: 2.9774
- =====

AdaBoost

=====

Model performance for Training set

- r2_score: 0.8977
 - mean_squared_error: 5.4464
 - mean_absolute_error: 4.4668
-

Model performance for Test set

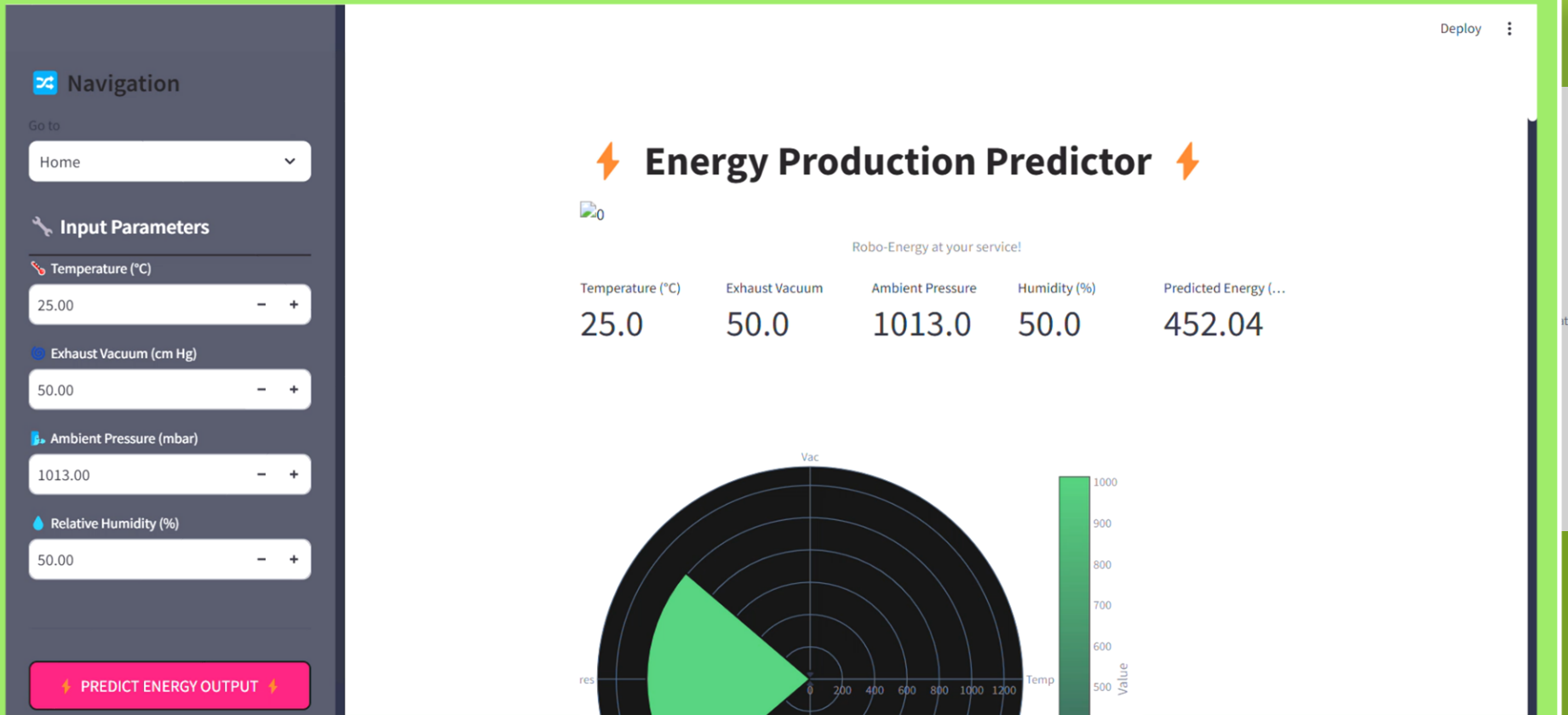
- r2_score: 0.8892
 - mean_squared_error: 5.6799
 - mean_absolute_error: 4.5934
- =====

Application Structure and Pages

Main Sections:

- **Home / Predictor:**
 - Inputs: Temperature, Exhaust Vacuum, Ambient Pressure, and Humidity.
 - Outputs: Predicted energy production, metrics visualization, and interactive charts (polar chart, gauge indicator, and sensitivity analysis).
- **About Project:**
 - Provides detailed information about the project methodology, key findings, and a typewriter-style description of the energy production model.
 - Uses an image related to powerplants to reinforce the theme.
- **More Insights:**
 - Showcases deeper analytical insights with responsive cards and interactive trend analysis charts.
 - Focuses on exploring energy production trends and data analytics.
- **AI Workflow:**
 - Outlines an eight-step process for website creation using AI tools such as ChatGPT and WordPress.
 - Displays relevant images dynamically pulled from Unsplash based on tailored keywords (e.g., “domain”, “wordpress”, “brainstorming,ai”).

Result



More Insights page

Navigation

Go to

More Insights

Explore the project workflow and additional analysis.



Project Workflow & Methodology

This section highlights our overall project workflow diagram, as well as key data visualizations used during the exploratory data analysis phase.



AI Workflow page

Deploy 

Navigation

Go to

AI Workflow 

Explore the ML development workflow and energy-specific resources.

ML Pipeline for Energy Production Prediction

Data Collection & Sources → 



Core Resources

 [UCI Energy Dataset](#) |  [IEA Data Services](#) |  [Kaggle Energy Datasets](#)

Feature Engineering → 

Technical Guides

 [Scikit-learn Feature Selection](#) |  [Feature Engineering Best Practices](#) |  [Pandas Documentation](#)

Model Development → 

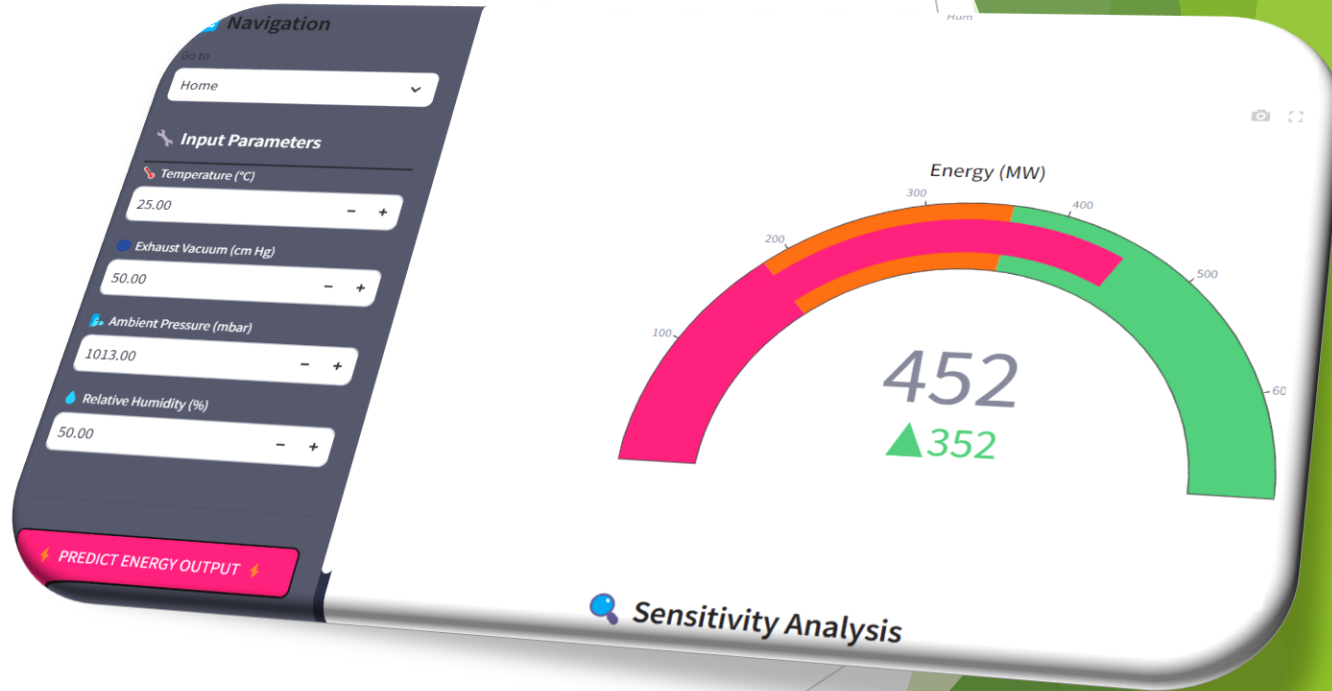
ML Frameworks

- [XGBoost Docs](#)
- [Scikit-learn Regression](#)
- [Joblib Serialization](#)

Validation & Testing → 

Final output

The Energy Production Prediction App built with Streamlit leverages a Random Forest model to accurately forecast energy production based on environmental factors like temperature, exhaust vacuum, ambient pressure, and relative humidity. This user-friendly and interactive tool provides valuable insights for optimizing energy systems, helping industries enhance efficiency and adapt to varying conditions effectively.



Conclusion:

- ↓ Based on the analysis, Random Forest, XGBoost, and GradientBoosting appear to be the top-performing models. They consistently achieve high R^2 -scores and low error metrics on both training and testing sets, suggesting they are well-suited for the given task.
- ↓ R^2 Score: Measures how well the model explains the variance in the data. A higher value (closer to 1) means better performance. Mean Absolute Error (MAE): The average of absolute differences between predicted and actual values. Lower is better. Mean Squared Error (MSE) and Root Mean Squared Error (RMSE): Penalizes larger errors more. Lower values indicate better predictions. Performance Comparison:
- ↓ Linear Regression Training set: R^2 : 0.9284 (Explains ~92.8% of the variance in the training data) MAE: 3.6213
Test set: R^2 : 0.9283 (Good generalization, close to training set performance) MAE: 3.6441 Linear Regression performs consistently well on both sets, showing no signs of overfitting.
- ↓ Random Forest Training set: R^2 : 0.9945 (Very high accuracy on training data) MAE: 0.8852 Test set: R^2 : 0.9617
(Still performs well but slightly drops on test set) MAE: 2.3406 Random Forest shows excellent performance but may slightly overfit the training data as the test performance is lower.

- DecisionTreeRegressor Training set: R^2 : 1.0000 (Perfect fit, likely overfitting) MAE: 0.0000 (No errors on the training set) Test set: R^2 : 0.9327 (Significant drop in test performance) MAE: 2.9695 This model is likely overfitting, as it performs perfectly on the training set but much worse on the test set.
- BaggingRegressor Training set: R^2 : 0.9923 (High accuracy on training set) MAE: 0.9987 Test set: R^2 : 0.9588 (Good generalization but slightly lower than training) MAE: 2.4380 Bagging performs similarly to Random Forest, with slightly less test accuracy, but good overall results.
- XGBoost Training set: R^2 : 0.9878 (Very high accuracy) MAE: 1.3643 Test set: R^2 : 0.9647 (Great generalization, slight drop compared to training) MAE: 2.2316 XGBoost performs very well on both sets and generalizes better than Decision Trees.
- AdaBoost Training set: R^2 : 0.8977 (Lower accuracy than other models on training) MAE: 4.4668 Test set: R^2 : 0.8892 (Similar test performance, but not as strong as others) MAE: 4.5934 AdaBoost performs consistently, but its R^2 and MAE values suggest it may not be the best choice for this dataset.
- GradientBoosting Training set: R^2 : 0.9539 (Strong performance on the training set) MAE: 2.8113 Test set: R^2 : 0.9461 (Good generalization with only a slight drop) MAE: 2.9774 GradientBoosting performs well, slightly below XGBoost but still solid on both sets.